

TUGAS PENDAHULUAN MODUL 05

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Tugas Pendahuluan Modul 05 Algoritma dan Pemrograman Rekursif

```
1. Package main
   Import "Fmt"
   Func Fibonacci (n int) int {
       If n == 1 || n == 2 {
           return 1
       }
       return Fibonacci (n-1) + Fibonacci (n-2)
   }

   Func main () {
       Var num int
       Fmt.Scan (&num)
       Fmt.Println (Fibonacci (num))
   }

2. Package main
   Import "Fmt"
   Func FCB (a, b int) int {
       If b == 0 {
           return a
       }
       return FCB (a % b, b)
   }
```

```
Func main () {
   Var x, y int
   Fmt.Scan (&x, &y)
   Fmt.Println (F (x, y))
}

3. Package main
   Import "Fmt"
   Func Faktorial (n int) int {
       If n <= 1 {
           return 1
       }
       return n * Faktorial (n-1)
   }

   Func main () {
       Var num int
       Fmt.Scan (&num)
       Fmt.Println (Faktorial (num))
   }
```

1.

```
TP5.no1.go
1 package main
2
3 import "fmt"
4
5 func fibonacci(n int) int {
6     if n == 1 || n == 2 {
7         return 1
8     }
9     return fibonacci(n-1) + fibonacci(n-2)
10 }
11
12 func main() {
13     var num int
14     fmt.Scan(&num)
15     fmt.Println(fibonacci(num))
16 }
```

```
C:\Users\Lenovo\OneDrive\Documents\ALPRO PRAKTIKUM>go run TP5.no1.go
5
5

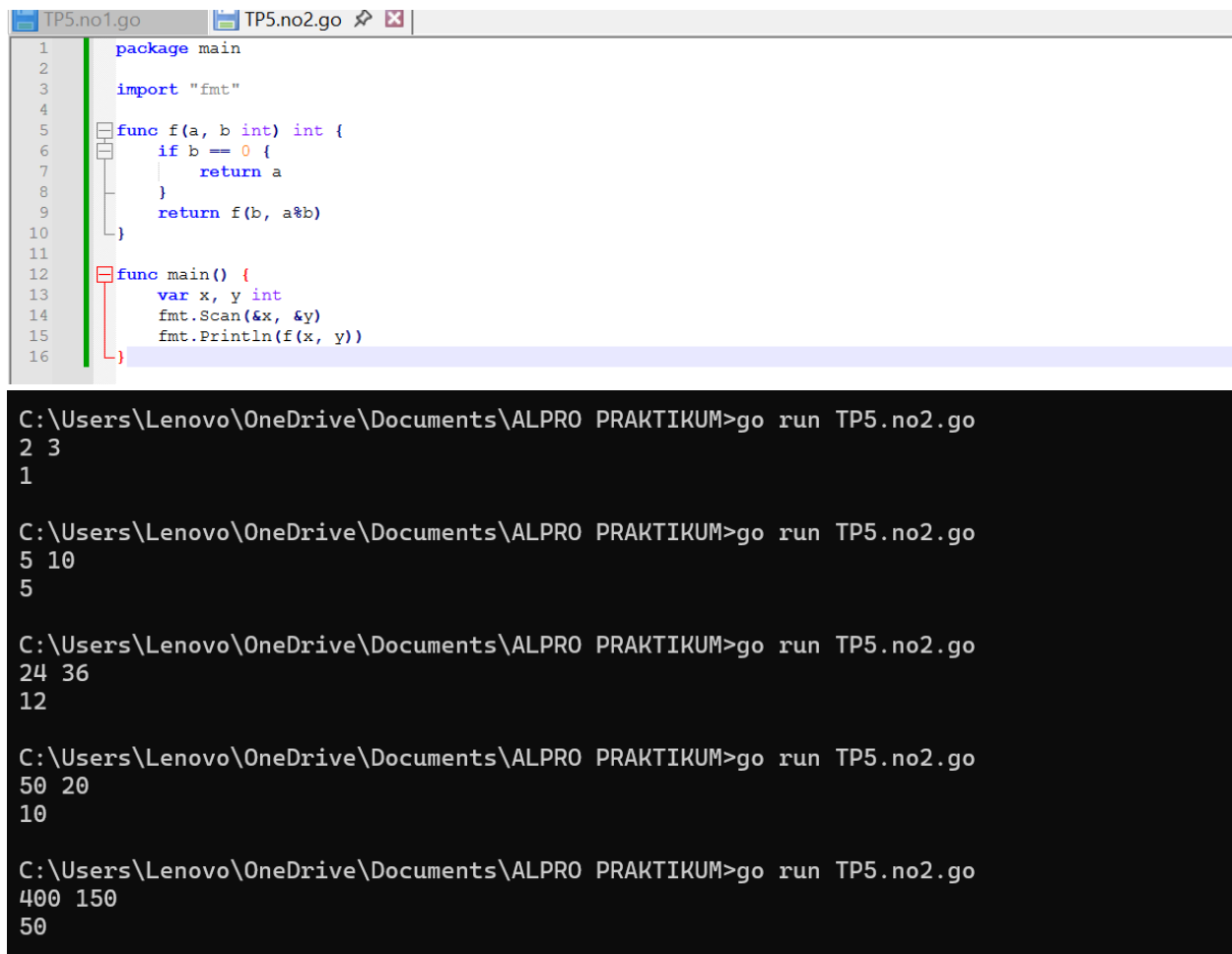
C:\Users\Lenovo\OneDrive\Documents\ALPRO PRAKTIKUM>go run TP5.no1.go
7
13

C:\Users\Lenovo\OneDrive\Documents\ALPRO PRAKTIKUM>go run TP5.no1.go
10
55

C:\Users\Lenovo\OneDrive\Documents\ALPRO PRAKTIKUM>go run TP5.no1.go
12
144

C:\Users\Lenovo\OneDrive\Documents\ALPRO PRAKTIKUM>go run TP5.no1.go
15
610
```

2.



The image shows a Go IDE with two tabs: TP5.no1.go and TP5.no2.go. The active tab, TP5.no2.go, contains the following Go code:

```
1 package main
2
3 import "fmt"
4
5 func f(a, b int) int {
6     if b == 0 {
7         return a
8     }
9     return f(b, a*b)
10 }
11
12 func main() {
13     var x, y int
14     fmt.Scan(&x, &y)
15     fmt.Println(f(x, y))
16 }
```

Below the code editor, the command prompt shows the execution of the program with various inputs and their corresponding outputs:

```
C:\Users\Lenovo\OneDrive\Documents\ALPRO PRAKTIKUM>go run TP5.no2.go
2 3
1

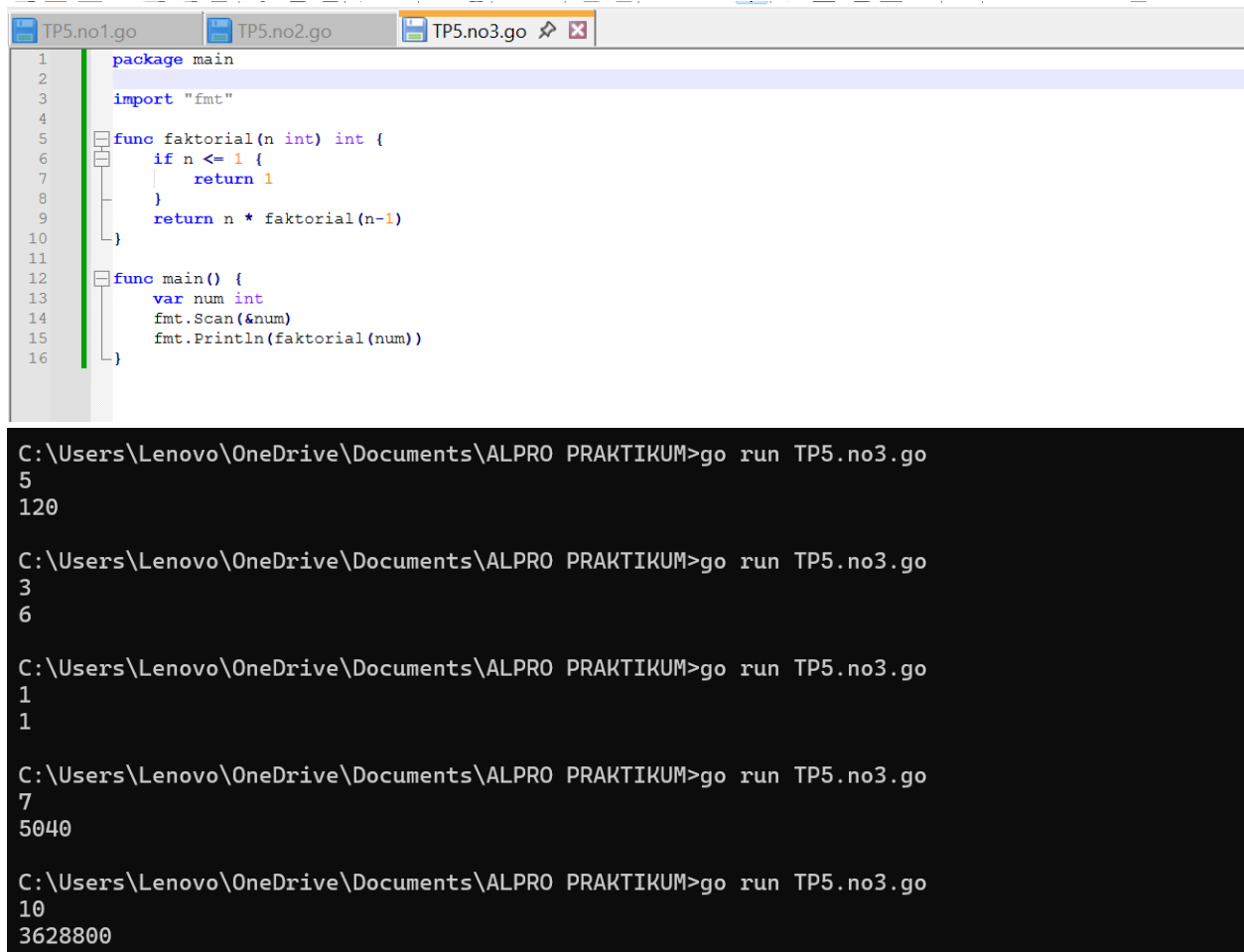
C:\Users\Lenovo\OneDrive\Documents\ALPRO PRAKTIKUM>go run TP5.no2.go
5 10
5

C:\Users\Lenovo\OneDrive\Documents\ALPRO PRAKTIKUM>go run TP5.no2.go
24 36
12

C:\Users\Lenovo\OneDrive\Documents\ALPRO PRAKTIKUM>go run TP5.no2.go
50 20
10

C:\Users\Lenovo\OneDrive\Documents\ALPRO PRAKTIKUM>go run TP5.no2.go
400 150
50
```

3.



The screenshot shows a Go IDE with three tabs: TP5.no1.go, TP5.no2.go, and TP5.no3.go. The active tab, TP5.no3.go, contains the following Go code:

```
1 package main
2
3 import "fmt"
4
5 func faktorial(n int) int {
6     if n <= 1 {
7         return 1
8     }
9     return n * faktorial(n-1)
10 }
11
12 func main() {
13     var num int
14     fmt.Scan(&num)
15     fmt.Println(faktorial(num))
16 }
```

Below the code editor, the command prompt shows the execution of the program for different input values:

```
C:\Users\Lenovo\OneDrive\Documents\ALPRO PRAKTIKUM>go run TP5.no3.go
5
120

C:\Users\Lenovo\OneDrive\Documents\ALPRO PRAKTIKUM>go run TP5.no3.go
3
6

C:\Users\Lenovo\OneDrive\Documents\ALPRO PRAKTIKUM>go run TP5.no3.go
1
1

C:\Users\Lenovo\OneDrive\Documents\ALPRO PRAKTIKUM>go run TP5.no3.go
7
5040

C:\Users\Lenovo\OneDrive\Documents\ALPRO PRAKTIKUM>go run TP5.no3.go
10
3628800
```